

Curriculum Vitae

Albertas Dvirnas

a. Contact Information

Address: Helena Elisabetsväg 2A, Umeå, Sweden
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b. Education

- **PhD in Theoretical Physics** (specialisation: Computational Biology)
Lund University, Aug 2017 - Jan 2022
Thesis: *Methods for barcode analysis in optical DNA mapping*
- **MSc in Applied Mathematics** (grade point average 10/10)
Kaunas University of Technology, Sep 2013 - Jun 2015
Thesis: *Analysis of the growth of algebras and their applicability in cryptography*
- **Erasmus exchange student**
Lund University, Aug 2014 - Jun 2015
- **BSc in Applied Mathematics** (grade point average 9.88/10)
Kaunas University of Technology, Sep 2009 - Jun 2013
Thesis: *Analysis of the applicability of unitary representations of the Artin's Braid Group in topological quantum computing*

c. Employment

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| • Project Assistant (100% research)
Department of Astronomy and Theoretical Physics, Lund University | Sep 2016 – Jul 2017 |
| • Summer Intern
Department of Astronomy and Theoretical Physics, Lund University | Jun 2016 – Aug 2016 |
| • Technical Assistant (Part time employment)
Department of Applied Mathematics, Kaunas University of Technology | Apr 2012 – Feb 2015 |

d. Leave of Absence

No periods of leave of absence or parental leave to report.

e. Postdoctoral Stays

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| • Postdoctoral Researcher
Department of Mathematics and Mathematical Statistics, Umeå University | October 2025 - current |
| • Postdoctoral Researcher
Department of Life Sciences, Chalmers University of Technology | Apr 2024 – Sep 2024 |
| • Postdoctoral Researcher
Computational Science for Health and Environment,
Centre for Environmental and Climate Science, Lund University | Mar 2022 – Feb 2024 |

f. Awards and Distinctions

- MSc degree awarded with distinction, Kaunas University of Technology, 2015
- BSc degree awarded with distinction, Kaunas University of Technology, 2013

g. Important Assignments

Reviewer for BMC Genomics

h. Research Grants

- Wenner-Gren Travel Grant (10,000 SEK), Wenner-Gren Foundations, 2026
- Travel Grant for H2O Conference (1,500 DKK), Department of Mathematics, Aarhus University, 2026
- Postdoctoral scholarship funded by Carl Tryggers Foundation, 2025–2027 (672,000 SEK), PI: Åke Brännström
- PhD partially funded by European Union Horizon 2020: “Genomic Diagnostics Beyond the Sequence

– BeyondSeq", PI: Yuval Ebenstein

i. References

Tobias Ambjörnsson (PhD supervisor)

Department of Earth and Environmental Sciences

tobias.ambjornsson@mgeo.lu.se

Fredrik Westerlund (Postdoc supervisor)

Department of Life Sciences, Chalmers University of Technology

fredrik.westerlund@chalmers.se

Åke Brännström (Current postdoc supervisor)

Department of Mathematics and Mathematical Statistics, Umeå University

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Per Stenberg (Current postdoc supervisor)

Department of Ecology, Environment and Geoscience, Umeå University

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Yuval Ebenstein (International collaborator)

Department of Biomedical Engineering, Tel Aviv University

uv@post.tau.ac.il

Eligijus Sakalauskas (Master thesis supervisor)

Kaunas University of Technology, Lithuania

eligijus.sakalauskas@ktu.lt

j. Other Relevant Information

Publicly Available Software

- **EMCCD-PIA** Electron-Multiplying Charge-Coupled Device Photon-optimal Image Analysis [Zenodo] [GitHub] (31 downloads, 143 views)
- **HCA** Chromosome alignment tool [Zenodo] [GitHub] (12 downloads, 152 views)
- **LLDEV** Optical mapping image processing [Zenodo] [GitHub] (17 downloads, 166 views)
- **SD3** Stained DNA Dot Detection [Zenodo] [GitHub] (41 downloads, 174 views)
- **DOGMA** De novo assembly of Densely labelled Optical DNA Maps using a matrix profile Approach [Zenodo] [GitHub]
- **HMMSV** Hidden Markov Model for Structural Variations [GitLab]

Teaching experience:

- Teaching assistant, Computational Physics course, Lund University, 2018–2020
- Course in Teaching and Learning in Higher Education (4.5 HP), 2019

Collaborators:

Active and planned collaborations with research groups at Lund University (SE), Chalmers University of Technology (SE), Umeå University (SE), Tel Aviv University (IL), Karolinska Institutet (SE), Uppsala University (SE), University of Zurich (CH), and UC Santa Cruz (US).

Selected talks:

- (Upcoming) July 21-26, 2026, "Diffusion map based forecasting from eDNA abundance time series", MMEE26, Cork, Ireland
- June 1, 2026, Bridging Matrix Profiles and Empirical Dynamic Modelling in Predictions for Environmental Data, Nordstat 2026, Helsinki, Finland
- May 4, 2026, "Footprints in time series", invited talk at Department of Earth and Environmental Sciences, Lund, Sweden
- January 19-23, 2026, "Empirical modeling perspective to ecosystem change", invited talk at Arctic Meeting for Adaptive Mechanisms in Biological Systems, Abisko, Sweden
- October 28, 2021, "Methods for barcode analysis in optical DNA mapping", invited talk at Nordic Computational Biology
- August 21, 2021, "Using Statistical Averages to Improve Similarity Search of Large DNA Molecules to the Human Genome", Joint Statistical Meetings 2021
- December 27, 2019, "Matricos profiliu paremtas motyvų atradimas laiko eilutėse", LJMS 9, Lithuania
- April 17, 2017, invited talk "DNA melting: theory vs. experiments" at Symposium Nanoconfined DNA, University of Gothenburg.
- July 3, 2014, contributed talk "Braid Group Cryptography", YTM 2014, Copenhagen, Denmark